# The Anisotropic Elasticity of Snow

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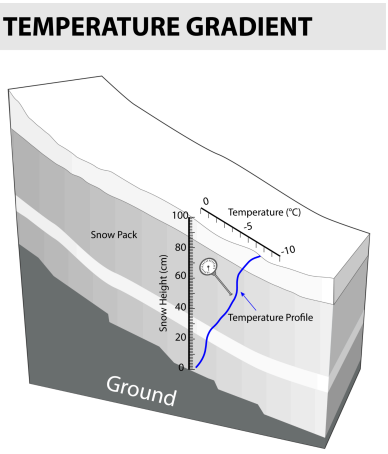
## Background

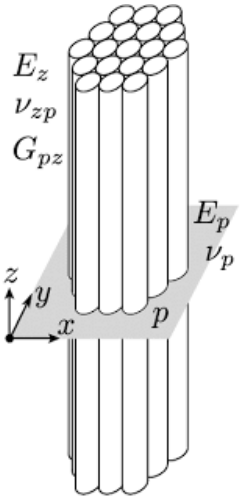
Snow and glaciers play an important role in Earth's climate system, water resources, and natural hazards. The frozen water that makes up large portions of glaciers and ice sheets around the world have huge variances in their material properties. The way these materials deform and transmit stress affects how glaciers respond to environmental forces and how seismic waves travel through them. Accurately predicting these behaviors requires knowledge of the elastic properties of snow, firn, and ice. Previous models have primarily focused on density as the controlling factor for stiffness, but many fail to account for anisotropy, or the directional dependence of material properties. Furthermore, no existing model accurately predicts elastic behavior across the full range of snow densities, from fresh snow to dense and non-airated glacial ice. The study presented in this paper addresses this limitation by developing a method to quantify the anisotropic elasticity of snow and incorporate the effects of microstructure into predictions of stiffness.

To understand the significance of this work, it is important to examine how snow evolves over time. Freshly fallen snow consists of a loose network of ice crystals separated by large volumes of air. Because of this porous structure, fresh snow has a very low ice volume fraction ($\phi$), typically between 0.02 and 0.05. As additional snowfall accumulates and the snowpack experiences various environmental factors including rain events, temperature variance, and pressure from additional snowfall, the layers of the snowpack compress. Ice grains become more tightly packed, bonds between grains strengthen, and air pockets gradually decrease in size. Through this process, snow transforms into firn, an intermediate material that contains both significant ice and air. Firn is categorized as snow that has persisted for a minimum of a year but has not yet compacted into glacial ice. Firn typically has an ice volume fraction between 0.4 and 0.8. Continued compaction and time eventually produces glacier ice, which has an ice volume fraction approaching 1.0 and contains relatively little pore space. This gradual transition from snow to firn and finally to glacier ice creates substantial variation in density and mechanical properties throughout a glacier.

The layered structure of a glacial snowpack is similar in many ways to that of a seasonal snowpack. In a typical winter snowpack, individual storms, wind events, and temperature changes produce layers with distinct characteristics that can often be seen visually or felt by hand. Some layers are soft and highly porous, while others are dense and resistant to deformation. These differences largely result from variations in the amount of trapped air and the degree of grain bonding within each layer. A glacier's accumulation zone experiences similar layering, although the layers become increasingly compressed as they are buried beneath new snowfall. Near the surface, the snowpack contains significant amounts of air and exhibits considerable structural variability. With increasing depth, compaction reduces pore space and increases the connectivity of the ice matrix. Even after layers become less visually distinct, differences in density and microstructure continue to influence the mechanical behavior of the material.

One of the most important characteristics of snow is anisotropy. Unlike an isotropic material, which behaves the same in every direction, snow develops directional properties as it settles under gravity and experiences temperature gradients. These processes encourage the formation of preferred grain orientations and elongated ice structures. As a result, snow is generally stiffer when loaded vertically than horizontally. This behavior is often modeled as transverse isotropy, a simplification that assumes material properties are identical in all horizontal directions but differ in the vertical direction. Near the surface, anisotropy is primarily controlled by the geometric arrangement of air pockets and ice bonds. Deeper within the snowpack, as density increases and air space decreases, the crystallographic orientation of ice crystals becomes increasingly important.




The elastic response of snow can be described using Hooke's Law, which relates stress and strain through a set of elastic constants. While a completely anisotropic material requires twenty-one independent elastic constants, the assumption of transverse isotropy reduces this requirement to five key elastic moduli. These parameters describe how the material responds to forces applied in different directions and help determine the speed at which seismic waves travel through the snowpack. Because seismic wave velocities vary between vertical and horizontal directions, measurements of elasticity provide valuable information about both density and internal structure. Consequently, understanding the relationship between density, anisotropy, and elasticity is essential for accurately modeling snow, firn, and glacier ice. This study builds upon that need by developing a model that incorporates both density and microstructural anisotropy, providing a more complete description of the mechanical behavior of snow across its full range of densities.

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{13} & 0 & 0 & 0 \\ C_{12} & C_{11} & C_{13} & 0 & 0 & 0 \\ C_{13} & C_{13} & C_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2}(C_{11} - C_{12}) \end{bmatrix}.$$

TI Elasticity Matrix

# Methodology

There were four key steps in the methodology of this experiment: data collection, number extraction, FEM analysis, and curve fitting.

Data Collection

Various snow samples were collected using μCT scans. This type of scan is a non-invasive scan and results in a detailed 3D model composed of voxels, or volumetric pixels. Each element on the grid has a specific ice volume fraction, Φ. This parameter specifies how much of each voxel is ice and how much is air. There were 391 samples collected. The samples were lab grown, collected from the Alps, Greenland, and Antarctica. In addition, the samples collected were all cubic, and across a full range of porosity.

Number Extraction

The first parameter needed was the ice volume fraction, which was determined from the μCT scan. The next piece needed was the geometric anisotropy parameter. This was determined by using a correlation function. This function determined the probability of there being ice in any direction by step size *r* if one also starts on ice. If the decay in the function is slow in the vertical direction and fast in the horizontal, then this means there are vertically elongated structures in the snow. If the decay is the opposite, then there are horizontal structures. This correlation function is a direct fingerprint of the microstructure's geometric anisotropy; it tells one how the fabric differs between directions. All the information from the correlation function gets put into a tensor for each sample. This means that each of the 391 samples are described by two parameters: ice volume fraction (Φ) and the geometric anisotropy parameter (α).

FEM Analysis

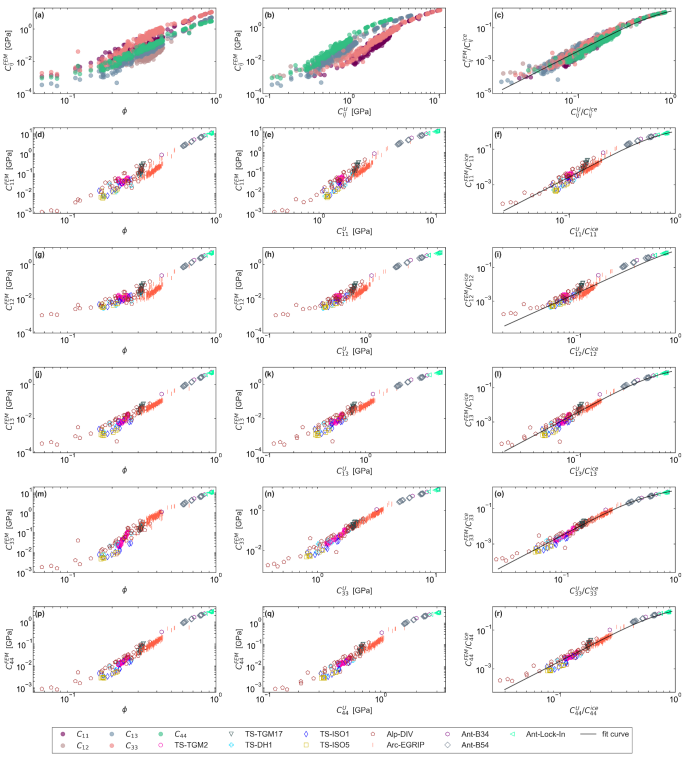
Once all the data has been collected FEM is used to compute the elasticity tensor for each sample. The 3d voxel map generated earlier by the scans is used directly as the mesh. Each voxel is assigned a bulk and shear modulus based off of established values from previous literature. In addition, the researchers made the standard assumption that the individual ice grains behave as an elastically isotropic polycrystalline ice matrix. This means that any macroscopic anisotropy is due to geometry, not the crystal alignment. This assumption is critical because it allows for the use of a specific boundary condition in the curve fitting step and without it, the math would not work. Essentially, if one were to zoom out slightly, the inconsistencies in the microstructure would "smooth out" and be negligible. Using FEM, a strain was applied to each sample 6 times: once in each independent direction (3 in each normal and 3 in each shear) to get the stress field for the whole sample. Following this, the volume average was calculated in order to get each column of the *C* matrix. This was performed for each sample, resulting in 391 elasticity tensors from Φ and α.

Curve Fitting

A special boundary condition known as the Hashim - Shtrikman bounds was used. The HS boundary condition already predicts a tensor from Φ and α, but it is an overestimation - it gives the theoretical maximum possible values. This means that the data needed to be normalized. The HS condition was used because it was accurate at each limit; when Φ approaches 0 and when Φ approaches 1. The shape of the curve is similar, but the magnitude is off. To correct this, the researchers performed a 2 parameter empirical transformation on top of the HS bounds to scale and shift their data so that it properly aligns, giving an accurate representation of the elasticity of snow from the volume fraction and geometric parameter.

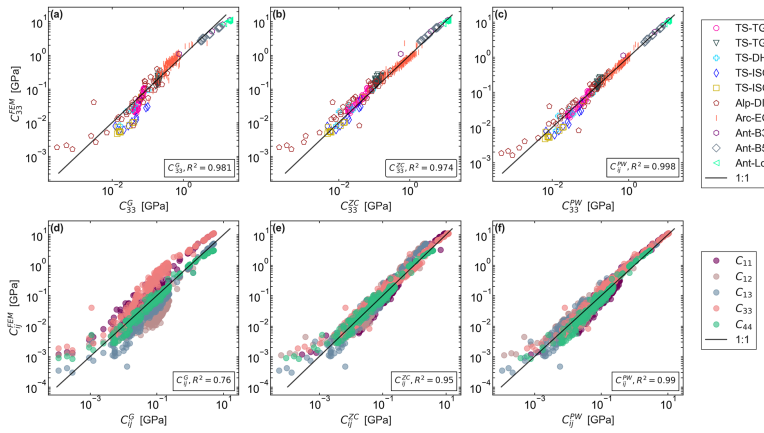
## Results & Discussion

In this study, finite-element simulations were performed on 391 micro-computed tomography (μCT) images representing snow, firn, and bubbly ice across a wide range of ice volume fractions (0.06–0.93) and geometrical anisotropies (α = 0.45–1.87). The simulations produced all five independent components of the transversely isotropic elasticity tensor, which were used to evaluate the relationship between elastic properties, density, and microstructural anisotropy.




The results showed that elastic stiffness increased strongly with ice volume fraction for all tensor components. When the simulated elastic moduli were plotted solely as a function of ice volume fraction, substantial scatter was observed, indicating that density alone could not fully explain variations in elastic behavior. This can be seen in the far left column of the figure. Incorporating the anisotropic Hashin–Shtrikman (HS) upper bound in the middle column significantly reduced this scatter, demonstrating that geometrical anisotropy contributes meaningfully to the elastic response of snow and firn. After normalization using the HS upper bound which is seen in the column on the far right of the figure, all elasticity tensor components collapsed onto a common trend that could be described using two fitting parameters per component. This shows that using both geometric anisotropy and ice volume fraction to calculate the elastic behavior is more accurate than using just the ice volume fraction alone, as in the density-based model of Gerling et al.

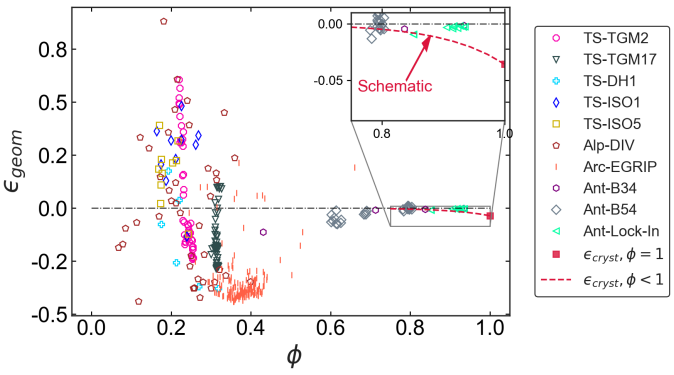
The proposed parameterization accurately reproduced the nonlinear increase in elastic stiffness from low-density snow to high-density bubbly ice. When we compare this new model with previous parameterizations, it showed significantly improved predictive performance. When fitted to all elasticity tensor components simultaneously, the new model achieved a coefficient of determination of $R^2 = 0.99$, compared with $R^2 = 0.952$ for the density-based model of Gerling et al. and $R^2 = 0.76$ for the Zysset–Curnier model. The largest improvements occurred for highly anisotropic samples, particularly those formed under strong temperature-gradients, which is to be expected since that is the main factor that this new model accounts for which the others did not.




At high ice volume fractions, the new proposed parameterization correctly approached the elastic modulus of pure ice and closely matched both finite-element simulations and theoretical predictions. Unlike previous density-based models, the

new formulation did not overestimate stiffness near the firn–ice transition and maintained physically realistic behavior across the entire density range.

Finally, the Thomsen anisotropy parameter was used to compare the influence of geometrical and crystallographic anisotropy. In comparison, the maximum crystallographic anisotropy is $\epsilon_{cryst} = -0.0356$, which is significantly less than the geometric anisotropy seen for samples with an ice volume fraction below 0.7. This means that the geometric anisotropy dominates the effect on the elastic behavior of the snow. However, as density increases toward pure ice, geometrical anisotropy decreases while crystallographic anisotropy becomes increasingly important. These findings suggest that both forms of anisotropy may contribute to the elastic properties of bubbly ice near the firn–ice transition. However, by just using geometric anisotropy we can still be extremely accurate in our predictions for firn where $0.3 < \phi < 0.7$.




## Summary

In conclusion, this study demonstrated that the elastic behavior of snow, firn, and bubbly ice is controlled not only by density but also by the geometric arrangement of the ice matrix. By combining ice volume fraction with a quantitative measurement of microstructural anisotropy, the researchers developed a new parameterization capable of more accurately predicting the five independent components of the transversely isotropic elasticity tensor across a wide range of snow densities. The finite-element simulations done showed that models based solely on density fail to capture important variations in stiffness, particularly in highly anisotropic snow where prediction errors can be large. The new model significantly reduced scatter in the data and achieved a coefficient of determination of $R = 0.99$, significantly outperforming previous density-based approaches.

The study shows the importance of geometric anisotropy in controlling the mechanical behavior of snow and firn, which was previously believed to be a much less significant factor. While crystallographic anisotropy becomes increasingly important as the material approaches pure ice, geometric anisotropy was found to dominate the elastic response through a significant portion of the snow to firn transition. These findings provide valuable insight into how snow deforms, transmits stress, and propagates seismic waves. Overall, the work covered in this paper represents a significant advancement in the modeling of snow and glacier materials and offers a more accurate framework for applications involving glacier dynamics, climate studies, snowpack stability, and geophysical monitoring of ice-covered environments. However, there is still work to be done in researching the upper end of the firn-ice transition zone, where crystal structure plays a greater role. In addition, these new measurements are yet to be tested in the field with seismic measurement tools which would help validate the applicability of this study.

## Team Contributions

**Saben:** Found and decided on the paper, created team contract, made slides for and presented results and analysis in presentation, wrote results section, did minor revising/editing.
**Wes:** Made background slides of presentation, and presented them. Also wrote background/intro of presentation and also kinda first drafted it.
**Arman:** Drafted presentation, made and presented methodology sections, wrote methodology section of essay.

## Originality Statement:

***"We confirm that our selected paper was not part of another course and was not previously used by any team member for another course project, presentation, or assignment. We understand that violations of this requirement may result in grade penalties at the professor's discretion."***

-Saben
-Wes
-Arman